



PennState

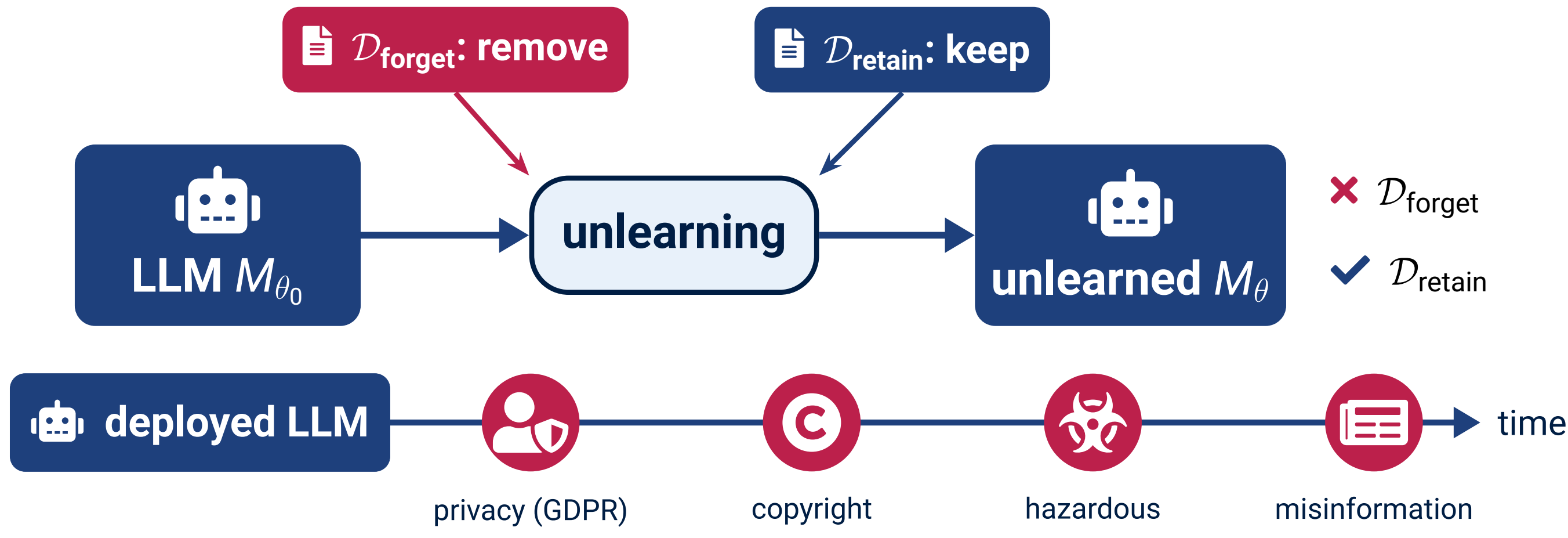
BLADE: Bilevel Low-rank Augmented-Lagrangian Erasure for LLM Unlearning

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Problem Overview



Objectives

Designing an unlearning method that is ...

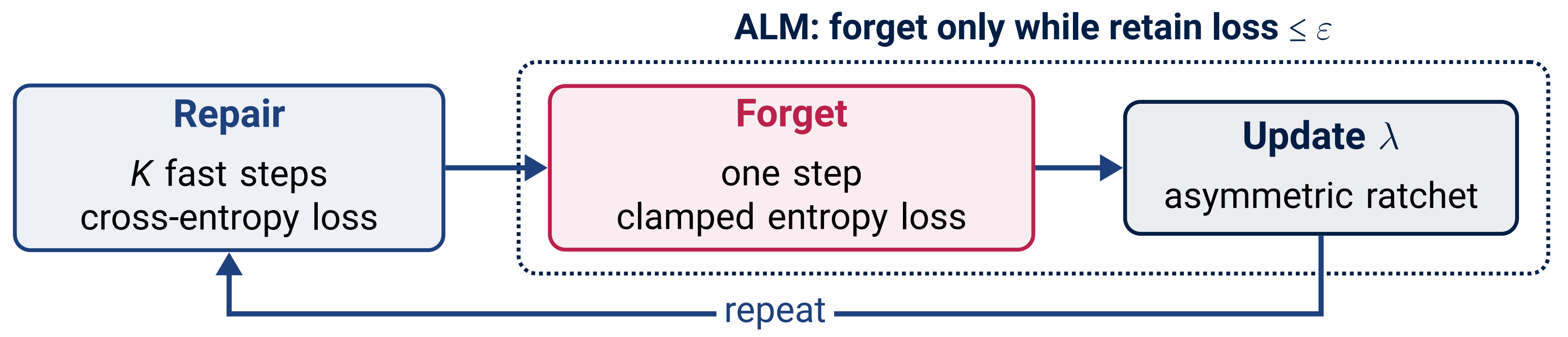
- Balanced**: Forgets with minimal model degradation
- Scalable**: Stays stable as forget set grows
- Sustainable**: Handles repeated unlearning requests
- Robust**: Resists extraction, jailbreak and membership attacks
- Efficient**: Inexpensive finetuning with smaller memory footprint

BLADE at a glance

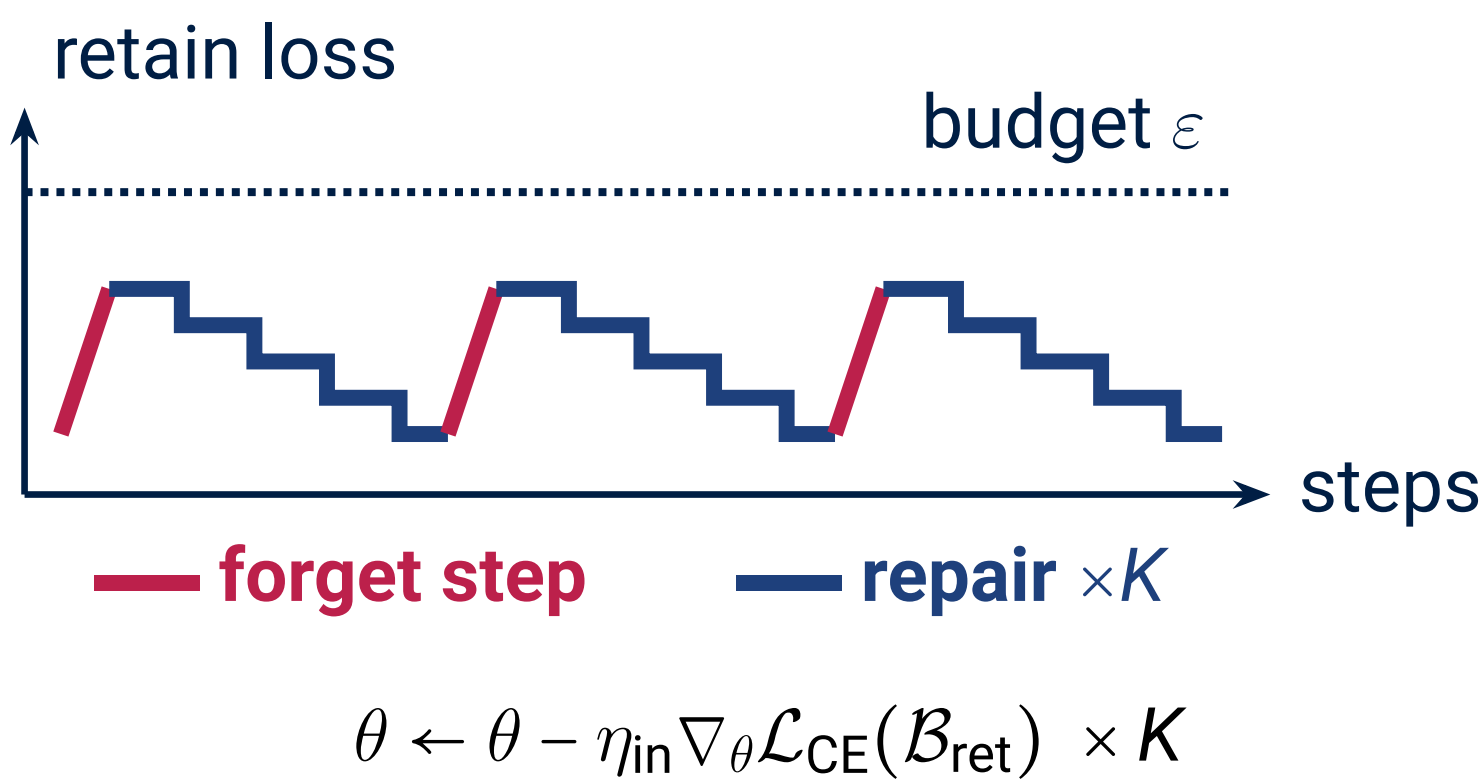
Formulation

- Forget** $\min_{\theta} \mathcal{L}_{\text{fgt}}(\theta^*; \mathcal{D}_{\text{forget}})$
- Retain budget** s.t. $\mathcal{L}_{\text{CE}}(\theta^*; \mathcal{D}_{\text{retain}}) \leq \varepsilon$
- Repair** $\theta^* \approx \arg \min_{\theta} \mathcal{L}_{\text{CE}}(\theta; \mathcal{D}_{\text{retain}})$
- LoRA** $W' = W_0 + \frac{\alpha}{r}BA$, W_0 frozen

One training step

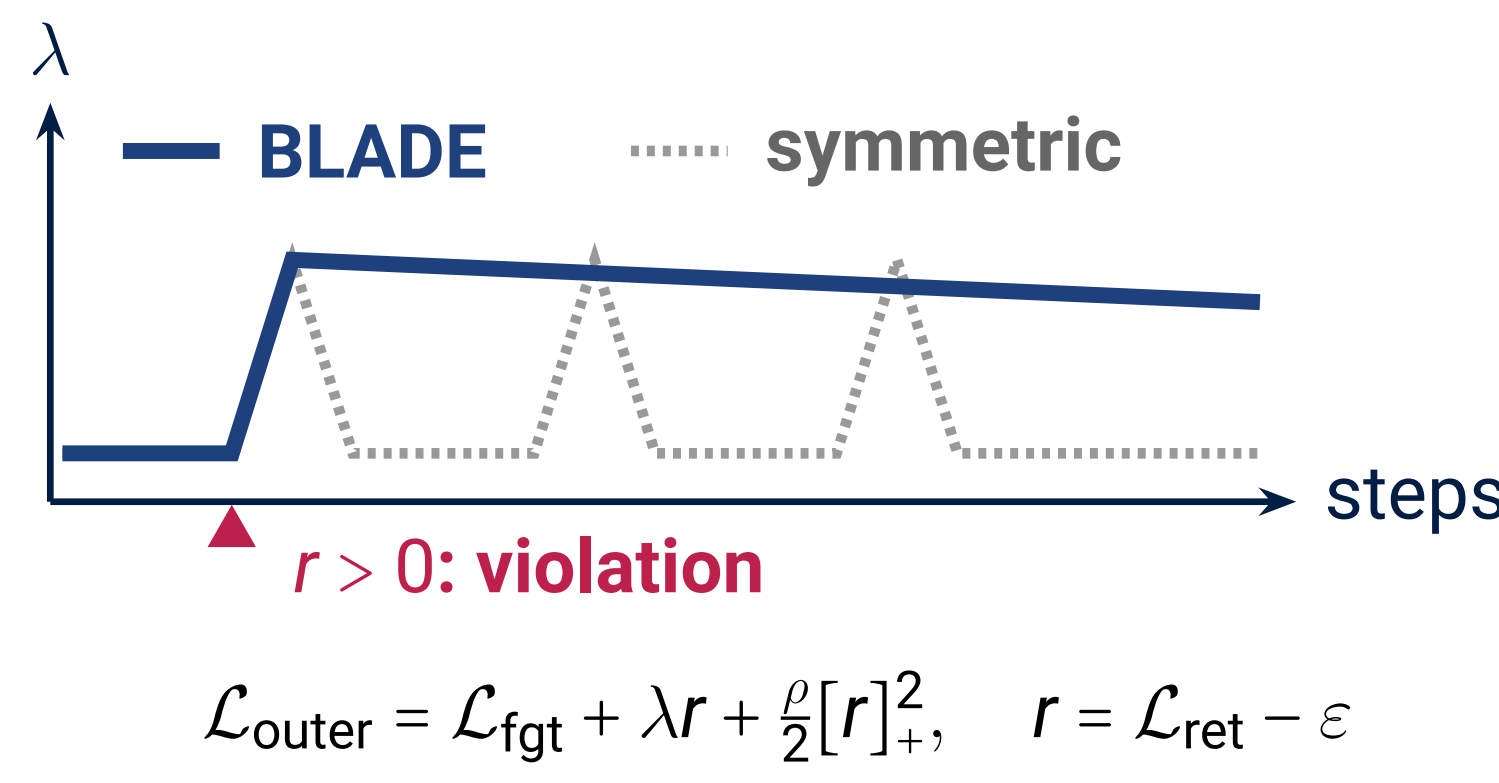


Repair-first principle



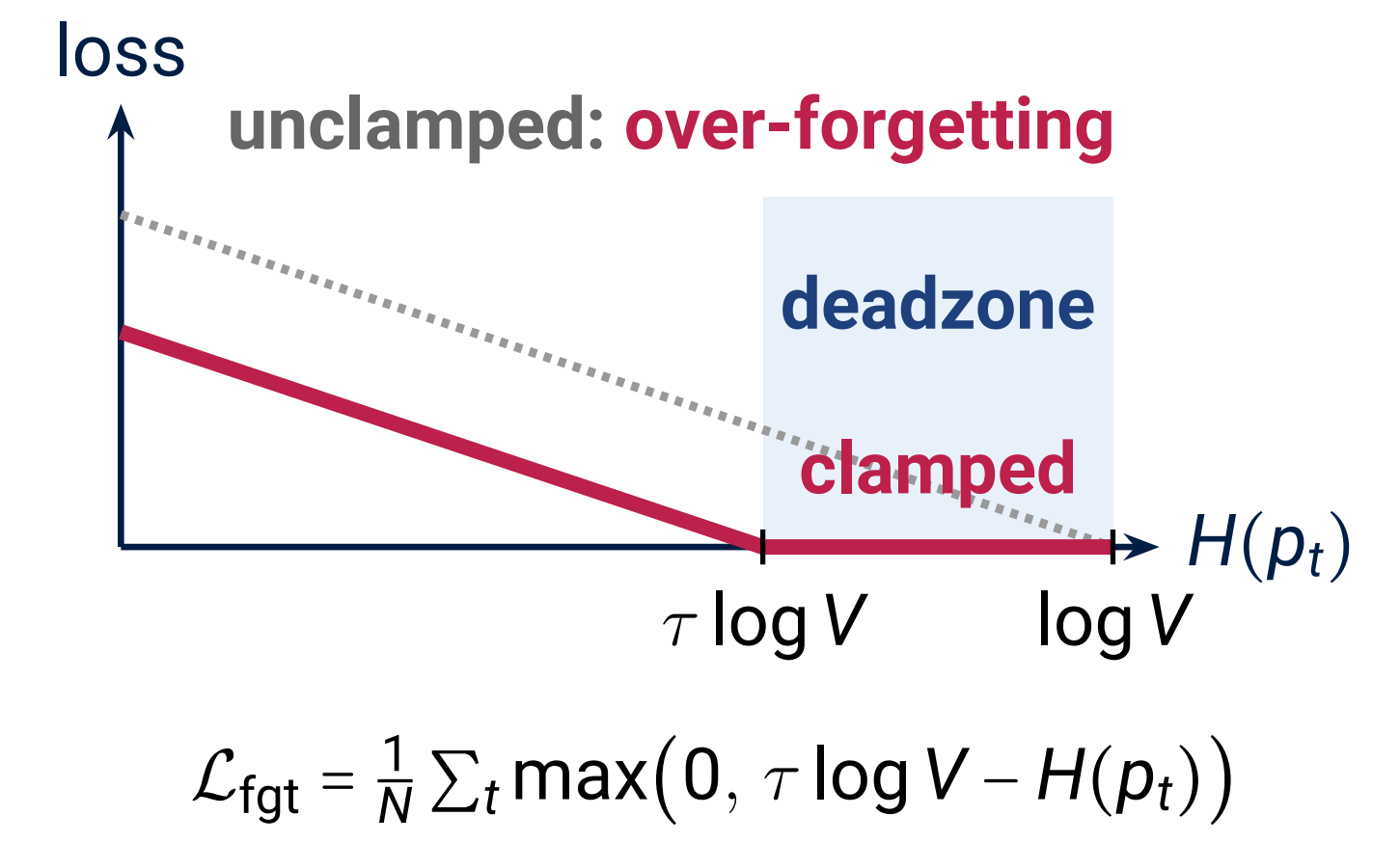
- Forget-retain **entanglement**
- Forget is **destructive** by nature
- Anchor** forget by retain

Asymmetric ALM



- Constrained** forget under retain
- λ grows fast, decays slowly

Clamped entropy



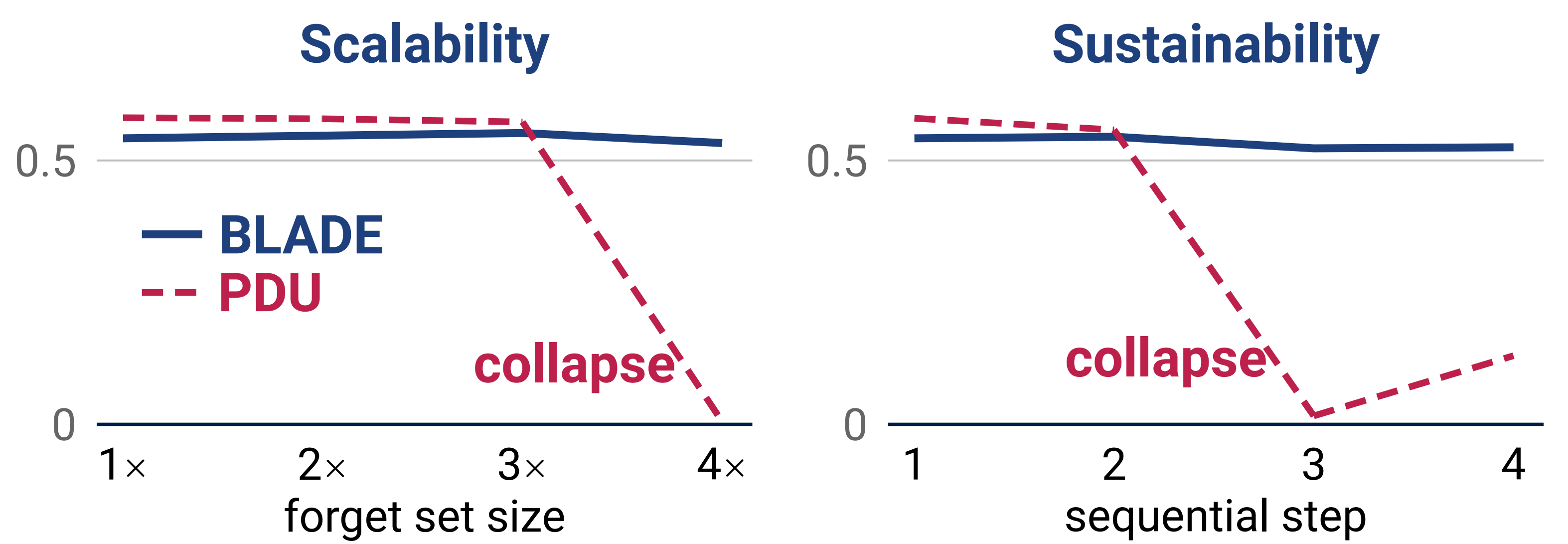
- $\nabla = 0$ **once forgotten**
- Bounded** and reference-free

Experiments

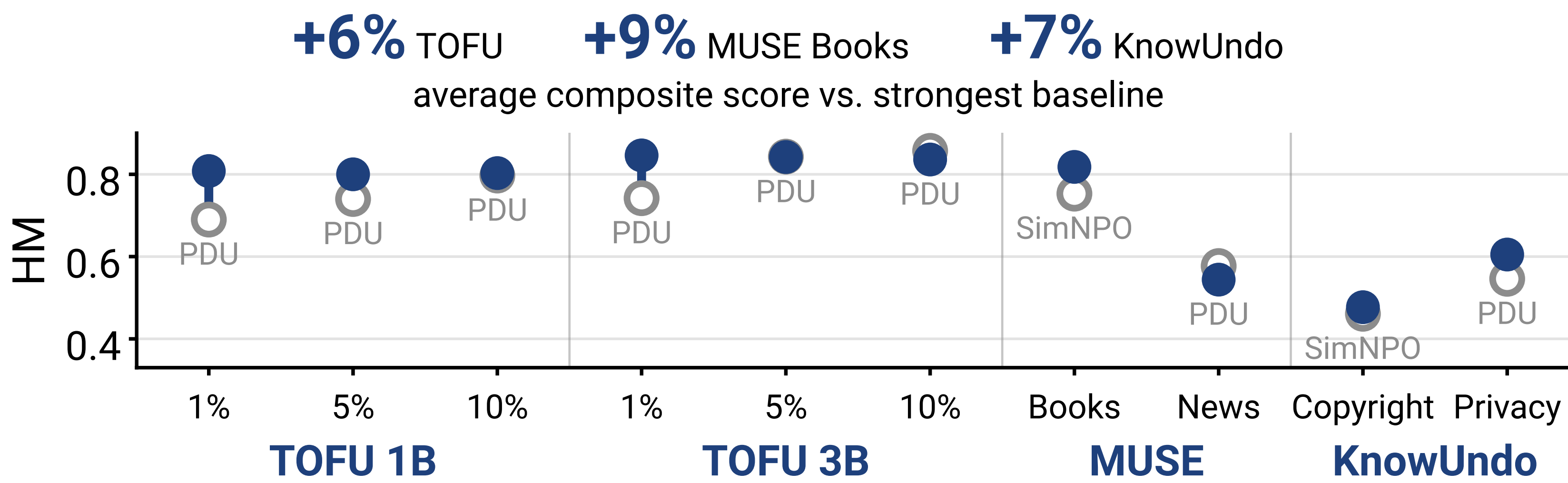
Setup

- Datasets**
- TOFU**: synthetic authors, Llama-3.2 1B/3B
 - MUSE**: Books, News, Llama-2-7B
 - KnowUndo**: copyright, privacy, Llama-2-7B-chat
- Baselines**
- GradAscent³, GradDiff, NPO⁵, SimNPO², RMU, BLURNPO⁴, PDU¹
- Metric**: HM of forget & retain axes

Key result: stable under scaling and repeated unlearning



Best on three benchmark families



Robust and efficient

- Extraction**: best adversarial HM
- Jailbreak**: lowest attack success
- MIA**: no leakage
- GCG**: lowest attack success
- Efficient**: 2–5× less GPU memory

¹. T. Entesari et al. "Constrained Entropic Unlearning: A Primal-Dual Framework for Large Language Models". 2025.

². C. Fan et al. "Simplicity Prevails: Rethinking Negative Preference Optimization for LLM Unlearning". 2025.

³. J. Jang et al. "Knowledge Unlearning for Mitigating Privacy Risks in Language Models". 2023.

⁴. H. Reiszadeh et al. "BLUR: A Bi-Level Optimization Approach for LLM Unlearning". 2026.

⁵. R. Zhang et al. "Negative Preference Optimization: From Catastrophic Collapse to Effective Unlearning". 2024.